SemiFT: Bridging the Gap between Vision Foundation Models and Remote Sensing Segmentation via Semi-supervised Adaptation

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Abstract

Parameter-Efficient Fine-Tuning (PEFT) has emerged as the mainstream for adapting Vision Foundation Models (VFMs) to downstream tasks. However, remote sensing (RS) semantic segmentation is typically constrained by limited annotations, under which conventional PEFT struggles to model complex spatial context of geospatial objects, limiting VFMs transferability. To enable efficient adaptation under label scarcity, we propose SemiFT, a novel semi-supervised fine-tuning framework for RS semantic segmentation. SemiFT introduces a Scale-Aware Mixture-of-Experts (SA-MoE) module that decouples the adaptation process into a dynamic collaboration among multiple scale-specific experts, improving multi-scale representation learning under insufficient supervision. We further leverage geometric priors from unlabeled data via a rotation-based geometric-invariant consistency regularization, which promotes orientation-robust predictions with limited labels. Experiments show that SemiFT demonstrates consistent gains in low-annotation scenarios with fewer trainable parameters, achieving an excellent accuracy–efficiency trade-off.

CCS Concepts

• Computing methodologies → Image segmentation; **• Applied computing → Earth and atmospheric sciences**.

Keywords

semi-supervised learning, parameter-efficient fine-tuning, vision foundation models, remote sensing semantic segmentation, mixture of experts

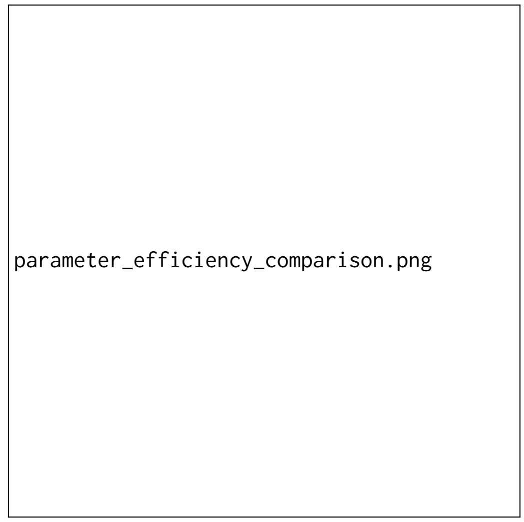

Figure 1: Efficiency-accuracy comparison of semi-supervised methods on Potsdam. The x-axis shows training time per epoch (s), y-axis represents mIoU (%), and bubble size indicates GPU memory (MB). SemiFT achieves the best trade-off with highest mIoU, lowest training time, and moderate memory usage.

1 Introduction

Vision Foundation Models (VFMs), empowered by large-scale pretraining, have learned robust representations and advanced the state of the art (SOTA) in remote sensing (RS) semantic segmentation [**? ? ?**]. However, practical deployment remains challenging. Pixel-level annotations are costly and scarce [**? ?**]. Moreover, fully fine-tuning VFMs requires prohibitive compute and memory resources [**? ?**]. These constraints hinder rapid iteration and scalable deployment across diverse regions and data sources, making efficient and affordable adaptation of VFMs a pressing priority [**? ?**].

Parameter-Efficient Fine-Tuning (PEFT) provides a natural starting point. By freezing most pretrained parameters and updating only a small set of lightweight modules, PEFT markedly reduces training and storage costs while retaining the general knowledge of VFMs, enabling fast adaptation across varied RS scenarios [**? ?**]. However, conventional PEFT typically relies on sufficient dense supervision to effectively fine-tune their modules. When annotation is limited, the lightweight modules being updated may not have enough representational capacity to capture the pronounced multi-scale variations and complex spatial structures of RS objects, thereby constraining adaptation effectiveness and generalization [**?**]. Meanwhile, unlabeled RS imagery is far easier to acquire and abundant at scale, providing a practical opportunity to mitigate the supervision bottleneck. Semi-supervised semantic segmentation (S4) provides a principled way to translate unlabeled data into additional training signals [**? ?**]. In practice, most S4 pipelines implicitly assume full-model or large-scale parameter updates to sustain effective consistency regularization or self-training. This motivates us to explore semi-supervised fine-tuning, where we aim to enhance parameter-efficient VFM adaptation with limited dense masks, leveraging sparse annotations and semi-supervised signals [**? ?**].

Semi-supervised fine-tuning faces the dual challenge of adapting the model with limited labeled data while preventing extensive parameter updates [**?**]. To address these, we propose SemiFT. SemiFT introduces a Scale-Aware Mixture-of-Experts (SA-MoE) module, which decouples adaptation into dynamic collaboration among multiple scale-specific experts, enhancing multi-scale representation learning with limited supervision. These lightweight experts, embedded into frozen Transformer layers and equipped with convolutional inductive biases [**? ?**], are selectively activated based on contextual scales, effectively modeling scale variations and spatial structures while keeping parameter usage minimal. Furthermore, we propose a rotation-based geometric-invariant consistency regularization (RGCR), which exploits geometric priors from unlabeled data to enforce consistent semantic predictions under rotations, thereby improving training stability and robustness in weakly supervised settings. As shown in Figure ??, SemiFT achieves the best accuracy-efficiency trade-off on the Potsdam dataset, delivering superior mIoU with the lowest training time and moderate memory consumption compared to existing methods. Extensive experiments across annotation-limited settings show that SemiFT improves accuracy with far fewer trainable parameters, achieving a strong accuracy–efficiency trade-off. This indicates semi-supervised fine-tuning as an effective pathway for leveraging VFMs in RS segmentation under persistent annotation scarcity. In summary, our contributions are as follows:

- We explore semi-supervised fine-tuning, a new adapting paradigm. It aims to transfer VFMs to downstream tasks with minimal trainable parameters while relying on only limited supervision.
- We propose SemiFT, a semi-supervised fine-tuning framework for RS semantic segmentation. SemiFT integrates a Scale-Aware Mixture-of-Experts (SA-MoE) module with lightweight, convolution-biased experts embedded into frozen Transformer layers and selectively activated by contextual scales, and further adopts rotation-based geometric-invariant consistency regularization to exploit RS priors and stabilize semi-supervised training.
- Extensive experiments across diverse settings validate the effectiveness of SemiFT.

2 Related Work

2.1 Semi-Supervised Remote Sensing Semantic Segmentation

Semi-supervised Learning (SSL) has become a cornerstone in remote sensing to mitigate the scarcity of pixel-level annotations [? ?]. Early frameworks such as Mean Teacher [?] and FixMatch[?] laid the groundwork by utilizing consistency regularization and pseudo-labeling. Recently, UniMatch [?] and its successor UniMatchv2 [?] have significantly elevated performance benchmarks by integrating robust pre-trained foundation models (e.g., DINOv2) [?], demonstrating that powerful feature priors are essential for low-data regimes.

To refine pseudo-label quality in semi-supervised remote sensing segmentation, various strategies have been explored. RanPaste [?] enhances reliability via a paste consistency mechanism that integrates ground truth labels into predictions. WSCL [?] leverages consistency between weakly and strongly augmented views, while SegMind [?] combines mean-teacher architecture with masked image modeling (MIM) and contrastive learning to improve spatial interactions and feature separability. CorrMatch [?] ensures coherent label propagation by modeling pairwise pixel similarities. To mitigate incorrect labels and class imbalance, DWL [?] introduces a decoupled weighting framework that adaptively weighs pseudo-labels based on confidence rankings. Addressing scale variations, ScaleMatch [?] utilizes a cross-scale interaction fusion (CIF) module to generate high-quality pseudo-labels through multi-scale prediction fusion and scale-invariant consistency. However, these frameworks mostly rely on full-parameter updates with heavy resource consumption and overlook rotational geometric priors inherent in RS imagery, leaving the potential of parameter-efficient adaptation for multi-scale representation under-explored.

2.2 Parameter-Efficient Fine-Tuning

As vision foundation models scale up, full-parameter fine-tuning becomes increasingly prohibitive for massive remote sensing datasets. Parameter-Efficient Fine-Tuning (PEFT) has emerged as a viable alternative, adapting models by tuning only a sparse subset of parameters [? ?]. Prompt-based methods, such as VPT [?], introduce learnable task-specific tokens, while selective tuning approaches like BitFit [?] and SSF [?] modulate existing weights through bias or scale adjustments.

Weight-based reparameterization and modular expansion represent the most prominent PEFT paradigms. LoRA [?] and its derivatives, such as FacT [?], parameterize weight updates via low-rank decomposition. To better capture domain-specific features, recent variants have integrated structural priors into the low-rank framework. For instance, Conv-LoRA [?] incorporates convolutional layers into the LoRA structure to introduce spatial inductive biases. Furthermore, the concept of Mixture-of-Experts (MoE) has been introduced to PEFT to handle feature heterogeneity. HydraLoRA [?] leverages an asymmetric MoE architecture with shared projections and multiple expert matrices to decouple complex features. Parallel to these, adapter-based methods, such as AdaptFormer [?], insert lightweight bottleneck modules into frozen backbones to achieve similar adaptation goals. However, existing PEFT paradigms, including recent MoE-based variants, mostly rely on fixed-scale transformations that fail to capture the vast scale variations of geospatial objects and lack the structural constraints needed to prevent overfitting to noisy pseudo-labels under sparse supervision.

3 Method

3.1 Preliminaries

In the semi-supervised semantic segmentation (S^4) for remote sensing images, we consider a labeled set $\mathcal{D}_l = \{(x_i^l, y_i^l)\}_{i=1}^{B_l}$ and a substantially larger unlabeled set $\mathcal{D}_u = \{x_j^u\}_{j=1}^{B_u}$, where $B_l \ll B_u$. Each image $x \in \mathbb{R}^{H\times W\times 3}$ is associated with a semantic map $y \in \mathbb{R}^{H\times W\times K}$, with K representing the number of land-cover categories. Our goal is to train a robust segmentation model $f(\cdot)$ by exploiting the structural priors in $\mathcal{D}_u$.

We adopt the weak-to-strong consistency framework as our base paradigm. For each unlabeled sample x_j^u, two views are generated: a weakly augmented view $x_j^{u_w} = \mathcal{A}_w(x_j^u)$ (e.g., flipping and cropping) and a strongly augmented view $x_j^{u_s} = \mathcal{A}_s(x_j^u)$ (e.g., color jittering and CutMix). The model $f(\cdot)$ generates prediction probability maps $p_j^{u_w}$ and $p_j^{u_s}$. A pseudo-label $\hat{y}_j$ is derived from the high-confidence regions of $p_j^{u_w}$ via $\hat{y}_j = \arg\max(p_j^{u_w})$, which then provides supervision for $p_j^{u_s}$. The joint optimization objective is formulated as:

$$\mathcal{L} = \mathcal{L}_s + \lambda_{u_s}\mathcal{L}_{u_s} \quad (1)$$

where λ_s denotes the standard cross-entropy loss on $\mathcal{D}_l$:

Fig2.pdf

Figure 2: Illustration of our SemiFT framework for semi-supervised remote sensing semantic segmentation. The Scale-Aware Mixture-of-Experts (SA-MoE) module is explicitly integrated into the encoder to capture multi-scale spatial representations. We propose Rotation-Geometric Consistency Regularization (RGCR) to enforce geometric invariance through rotation and scale transformations, while employing a teacher-student architecture with EMA updates to stabilize semi-supervised learning under label scarcity

$$\mathcal{L}_s = \frac{1}{B_l} \sum_{i=1}^{B_l} \ell_{ce}(y_i^l, f(x_i^l)) \tag{2}$$

The unsupervised term $\mathcal{L}_{u_s}$ enforces consistency while filtering out unreliable signals via confidence thresholding:

$$\mathcal{L}_{u_s} = \frac{1}{B_u}\sum_{j=1}^{B_u} \mathbf{1}(\max(p_j^{u_w}) \geq \tau)\ell_{ce}(\hat{y}_j, p_j^{u_s}) \tag{3}$$

Here, τ represents the confidence threshold, and $1(\cdot)$ is the indicator function that masks pixels with low prediction certainty to prevent error propagation during training.

3.2 SemiFT framework

Figure **??** illustrates the overall framework of SemiFT. Our method employs a dual-branch architecture for joint processing of labeled and unlabeled data, where a SA-MoE module is integrated into the encoder to enhance multi-scale feature extraction.

For each input image x_j, we employ an adaptive context sampling strategy to generate two complementary views: main view $x_j^{u_w}$ and context view $x_j^{u_s}$. The context view is further subjected to random scaling (s) and rotation (θ). To enforce geometric invariance, we apply inverse transformations-specifically inverse rotation ($-\theta$) and rescaling ($1/s$)-to the resulting predictions $p_j^{u_s}$, ensuring precise spatial alignment with the main view.

In the semi-supervised pipeline, the main view generates high-confidence pseudo-labels to supervise the context view via a consistency regularization $\mathcal{L}_u^{rgcr}$. For labeled data, both views are supervised by the GT, where the context view's supervision acts as an additional geometric-invariant signal $\mathcal{L}_x^{rgcr}$. The overall loss function can be formulated as:

$$\mathcal{L} = \mathcal{L}_s + \lambda_x \mathcal{L}_x^{rgcr} + \lambda_{u_s}\mathcal{L}_{u_s} + \lambda_{u_s}\mathcal{L}_u^{rgcr} + \mathcal{L}_{\text{load}} \tag{4}$$

where λ_x and λ_{u_s} represent the loss weights for the geometric-invariant consistency regularization and the semi-supervised consistency regularization on labeled and unlabeled data.

3.3 RGCR

In response to the challenge of orientation volatility in remote sensing imagery, we introduce the Rotation-Geometric Consistency Regularization (RGCR). By integrating unlabeled geometric priors, RGCR imposes semantic consistency constraints across various geometric perturbations to stabilize semi-supervised learning. This consistency is implemented through an adaptive context sampling strategy, which serves as the core mechanism for generating spatially aligned yet geometrically diverse views. The sampling process is primarily driven by two sets of parameters within the Geometric Transformations:

- Rotation Angles (θ): Angles are sampled from a uniform distribution $\theta \sim \mathcal{U}(0, 2\pi)$ to bolster the model's robustness against arbitrary object orientations.
- Scale Factor (s): Scaling ratios are selected from a discrete set $s \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$. This configuration effectively simulates a range of spatial resolutions, spanning from fine-grained textures to broad structural layouts.

From an original remote sensing image x, we randomly crop a main view x_m at position (x, y) and apply weak augmentation $\mathcal{A}_w(\cdot)$ to obtain x_m^w for pseudo-label generation. A context view x_c is then cropped from the same image at coordinates (x_c, y_c), computed according to the scale factor s:

$$(x_c, y_c) = \begin{cases} (x - \delta_s \cdot S, y - \delta_s \cdot S), & \text{if } s > 1.0 \\ (x + \delta_s \cdot S, y + \delta_s \cdot S), & \text{if } s < 1.0 \\ (x, y), & \text{if } s = 1.0 \end{cases} \tag{5}$$

where δ_s is a random offset that ensures spatial overlap between views. To ensure spatial overlap between the context view and the main view, the cropping regions satisfy the constraint:

$$|\mathcal{R}_m \cap \mathcal{R}_c| = \min(|\mathcal{R}_m|, |\mathcal{R}_c|) \tag{6}$$

where $\mathcal{R}_m = [x, x + size] \times [y, y + size]$ represents the main view region, and $\mathcal{R}_c = [x_c, x_c + s \times size] \times [y_c, y_c + s \times size]$ represents the context view region.

3.4 SA-MoE

The SA-MoE module is designed to capture multi-scale spatial representations through dynamic expert collaboration. As illustrated in Figure **??**, the SA-MoE processes the input X through three sequential stages: (1) Low-rank Projection, compressing the input into a compact latent space; (2) Scale-Aware Routing, where tokens are dynamically dispatched to dual-branch experts; and (3) Residual Restoration, which projects features back to the original dimension.

For an input tensor $X \in \mathbb{R}^{N \times d}$, we first compress the feature dimension into a low-rank subspace $r \ll d$ using a downsampling projection:

$$X' = \phi_{\text{down}}(X) = \text{GELU}(W_{\text{down}} X) \tag{7}$$

To incorporate global context, the gating network utilizes the class token X_{cls} to compute routing weights:

$$g = \text{Softmax}(W_g X_{\text{cls}} / \tau) \tag{8}$$

An auxiliary load-balancing loss $\mathcal{L}_{\text{load}}$ is employed to prevent expert collapse by minimizing the coefficient of variation of g.

Each expert in SemiFT is designed as a dual-branch convolutional structure to extract multi-scale spatial features. This design is motivated by the observation that different spatial frequencies capture complementary information: low-frequency components encode global context and coarse structures, while high-frequency components preserve fine-grained details and boundaries. For an expert with scale factor s_i, the processing involves two parallel paths:

- **Upsampling branch** captures fine-grained details by interpolating features to a higher resolution before applying Depthwise Convolution (DWConv);
- **Downsampling branch** aggregates broader context by processing features at a reduced resolution.

Specifically, given the input features X for an expert with scale factor s_i, the operations of these two branches can be formally expressed as:

$$\begin{aligned} \mathbf{X}_{up}^{(i)} &= \mathcal{D}_{1/s_i}(\text{DWConv}(\mathcal{U}_{s_i}(\mathbf{X}))) \\ \mathbf{X}_{down}^{(i)} &= \mathcal{U}_{s_i}(\text{DWConv}(\mathcal{D}_{1/s_i}(\mathbf{X}))) \end{aligned} \tag{9}$$

where $\mathcal{U}_\sigma(\cdot)$ and $\mathcal{D}_\sigma(\cdot)$ represent the bilinear interpolation-based upsampling and downsampling operations with a scaling factor σ, respectively. DWConv denotes depthwise separable convolution with kernel size 3×3, which provides efficient local feature fig3_1.png

Figure 3: Illustration of our Scale-Aware Mixture-of-Experts (SA-MoE) module for multi-scale feature extraction. The module is seamlessly integrated into frozen Transformer layers to capture diverse spatial representations. We propose a dual-branch convolutional expert design with up- and down-sampling paths to extract fine-grained and coarse-grained features, while employing a scale-aware routing mechanism to dynamically dispatch tokens based on global context, achieving parameter-efficient adaptation with sparse activation.

extraction with minimal parameter overhead. The use of depthwise convolution instead of standard convolution further reduces computational cost while preserving spatial inductive biases crucial for dense prediction tasks.

The outputs from both branches are combined through element-wise addition:

$$\mathrm{X}_{expert}^{(i)} = \mathrm{X}_{up}^{(i)} + \mathrm{X}_{down}^{(i)} \tag{10}$$

This dual-branch design enables each expert to simultaneously capture multi-scale information: the upsampling branch focuses on fine-grained details such as object boundaries and small structures, while the downsampling branch aggregates broader contextual patterns. By assigning different scale factors to different experts (e.g., $s_1 = 1, s_2 = 2, s_3 = 4, s_4 = 8$), SemiFT is capable of capturing fine-grained and coarse-grained features across diverse receptive fields, thereby effectively addressing multi-scale variations of target objects. The outputs from each expert are first aggregated via a weighted sum based on gating weights, then fused with the original input through a residual connection followed by normalization. To restore the feature dimensionality, the processed patch tokens are concatenated with global tokens ($X_{\mathrm{cls}}, X_{\mathrm{rel}}$) and projected back to the high-dimensional space $\mathbb{R}^d$ via an up-projection matrix W_{up}. The entire aggregation and restoration pipeline is formulated as:

$$X_{\text{combined}} = \sum_{i=1}^{n_{\text{experts}}} g_i \cdot X_{\text{expert}}^{(i)} \tag{11}$$

$$X'_{\text{patch}} = \text{LayerNorm}(X_{\text{patch}} + X_{\text{combined}}) \tag{12}$$

$$\tilde{X} = \text{LayerScale}\left(W_{\text{up}} \cdot \text{Concat}(X_{\text{cls}}, X_{\text{rel}}, X'_{\text{patch}})\right)) \tag{13}$$

where $W_{\text{up}} \in \mathbb{R}^{d \times r}$ is the up-projection matrix, and LayerScale introduces learnable scaling parameters to stabilize training in deep networks.

The parameter efficiency of SemiFT stems from several key design choices. First, the low-rank bottleneck reduces parameters from $O(d^2)$ to $O(dr)$. Second, each expert employs depthwise separable convolution, reducing parameters from $O(k^2C^2)$ to $O(k^2C + C^2)$. Third, the gating mechanism ensures sparse activation, providing computational savings during inference. Compared to full fine-tuning, SemiFT achieves 90% parameter savings while maintaining competitive performance.

4 Experiments

4.1 Datasets

Experiments are conducted on two standard remote sensing semantic segmentation benchmarks, Vaihingen and Potsdam, provided by ISPRS WG II/4 [?], to ensure authoritative and comparable evaluation. Vaihingen contains 33 urban aerial images with varying sizes (average $\approx 2494 \times 2064$ pixels) and a spatial resolution of 9 cm. Potsdam contains 38 images, each of size 6000×6000 pixels, with a spatial resolution of 5 cm. Both datasets share the same five land-cover classes: impervious surfaces, buildings, low vegetation, trees, and cars. For the semi-supervised setting, we conduct experiments using 1, 2, and 4 labeled images from the Vaihingen training set, and 1, 3, and 6 labeled images from the Potsdam training set, respectively; the remaining training images are treated as unlabeled. Evaluation is performed on the official test sets to assess model performance under extremely limited annotation budgets.

4.2 Implementation Details.

We adopt a UperNet segmentation model with DINOv3 as the backbone as our baseline. All experiments are implemented in PyTorch and trained on four NVIDIA RTX A6000 GPUs with an input resolution of 512×512. We use the AdamW optimizer with a weight decay of 0.01, a base learning rate of 1×10^{-4}, and the OneCycleLR learning rate scheduler. Models are trained for 75 epochs on both Vaihingen and Potsdam with a batch size of 4. The consistency loss weight and confidence threshold are both set to 0.5. All segmentation performance evaluations are reported in terms of mean Intersection over Union (mIoU).

4.3 Comparison with State-of-the-Art Methods.

In this part, we demonstrate the outstanding performance of SemiFT with comparison of state-of-the-art methods on both datasets under different partition protocols, including RanPaste [?], WSCL [?], SegMind [?], UniMatch [?], CorrMatch [?], DWL [?], ScaleMatch [?], and UniMatchV2 [?].

Method	Backbone	1/24	1/8	1/4
SupOnly	RN-101	67.19	78.79	81.79
RanPaste [TGRS'22]	RN-101	77.13	82.80	82.98
WSCL [TGRS'23]	RN-101	73.79	82.76	83.49
SegMind [TGRS'23]	RN-101	76.08	83.38	84.21
UniMatch [CVPR'23]	RN-101	66.63	83.22	83.47
CorrMatch [CVPR'24]	RN-101	65.28	82.87	83.51
DWL [ISPRS'24]	RN-101	75.58	82.70	83.47
ScaleMatch [AAAI'25]	RN-101	77.26	83.89	84.10
SupOnly	DINOv3-B	77.51	82.21	84.09
UniMatchV2 [TPAMI'25]	DINOv3-B	81.71	85.75	86.06
SemiFT	DINOv3-S	79.04	84.25	85.68
SemiFT	DINOv3-B	**81.99**	**86.81**	**86.58**

Table 1: Comparison with SOTAs on the Potsdam dataset.

Potsdam dataset. Table **??** reports the comparison with representative S4 methods under three label splits (1/24, 1/8, 1/4). Under the RN-101 backbone, semi-supervised baselines consistently outperform SupOnly, and ScaleMatch achieves the best performance across all splits. In particular, ScaleMatch improves over the strongest RN-101 competitor by +0.13, +0.67, and +0.59 on 1/24, 1/8, and 1/4, respectively, and brings a clear gain over SupOnly (e.g., +10.07 on 1/24). When upgrading the backbone to DINOv3-B, the performance further rises, showing the benefit of stronger visual priors. More importantly, SemiFT yields the best results among DINOv3-B based methods, surpassing UniMatchV2 by +1.34 / +0.39 / +0.45 on 1/24 / 1/8 / 1/4, which indicates that our method can effectively leverage larger-capacity backbones and remains robust under different label budgets.

Vaihingen dataset. Table **??** presents the results on Vaihingen with the same partition protocols. Similar to Potsdam, under RN-101, ScaleMatch remains the strongest baseline and consistently delivers top performance over prior S4 methods across splits, validating the effectiveness of the weak-to-strong style learning in urban aerial segmentation. When upgrading the backbone to DINOv3-B, SemiFT achieves the best performance among all methods, surpassing UniMatchV2 by +1.72, +1.45, and +2.45 on 1/16, 1/8, and 1/4, respectively. In particular, SemiFT significantly outperforms the supervised baseline SupOnly by +5.42, +3.51, and +3.56 on the three splits, demonstrating substantial improvements even under extremely limited supervision. Notably, SemiFT (DINOv3-S) also achieves competitive performance while using fewer parameters, further validating the parameter efficiency of our approach. Overall, the consistent and substantial improvements across both datasets suggest that SemiFT effectively leverages the scale-aware MoE architecture and geometric-invariant consistency to achieve robust performance in low-label regimes, making it a practical solution for semi-supervised remote sensing segmentation.

Method	Backbone	1/16	1/8	1/4
SupOnly	RN-101	68.43	71.89	74.38
RanPaste [TGRS'22]	RN-101	70.08	75.51	76.00
WSCL [TGRS'23]	RN-101	71.35	74.64	75.74
SegMind [TGRS'23]	RN-101	72.55	74.11	75.58
UniMatch [CVPR'23]	RN-101	67.77	69.31	77.02
CorrMatch [CVPR'24]	RN-101	68.43	75.78	76.80
DWL [ISPRS'24]	RN-101	72.09	74.80	76.26
ScaleMatch [AAAI'25]	RN-101	71.70	75.60	76.69
SupOnly	DINOv3-B	73.36	75.80	76.99
UniMatchV2 [TPAMI'25]	DINOv3-B	77.06	77.86	78.10
SemiFT	DINOv3-S	76.37	75.83	76.04
SemiFT	DINOv3-B	**78.78**	**80.16**	**80.94**

Table 2: Comparison with SOTAs on the Vaihingen dataset.

Baseline	RGCR	SA-MoE	Params	Mem.	1/16	1/8	1/4
✓	–	–	102.4M	11710 MB	78.29	78.25	78.68
✓	✓	–	102.4M	16539 MB	79.27	79.29	78.76
✓	✓	✓	10.38M	12907 MB	78.78	79.31	80.55

Table 3: Ablation of RGCR and SA-MoE. Params are trainable parameters; Mem. denotes peak GPU memory.

4.4 Ablation Studies

Component Analysis. RGCR and SA-MoE constitute the two core components of our SemiFT framework. We systematically evaluate their individual and combined effectiveness in Table **??** under three label splits (1/16, 1/8, and 1/4), reporting both segmentation accuracy and resource consumption. Adding RGCR on top of the baseline yields consistent gains across all splits, improving performance from 78.29/78.25/78.68 to 79.27/79.29/78.76, corresponding to improvements of +0.98, +1.04, and +0.08 mIoU points, respectively. This demonstrates that the proposed rotation-based geometric-invariant consistency regularization effectively enhances learning in low-label regimes by exploiting geometric priors from unlabeled data and enforcing robustness to rotational variations. Further incorporating SA-MoE achieves the best overall performance on the 1/8 and 1/4 splits (79.31 and 80.55 mIoU) while simultaneously reducing the number of trainable parameters from 102.4M to 10.38M—a substantial 90% reduction. Although the 1/16 split shows a slight performance dip (-0.49 mIoU), the overall results highlight the complementary nature of RGCR and SA-MoE: RGCR provides a strong performance boost via geometric consistency regularization, while SA-MoE improves parameter efficiency and further refines multi-scale representation learning with minimal computational overhead. The combination of these components enables SemiFT to achieve competitive performance with significantly fewer trainable parameters, demonstrating an excellent accuracy–efficiency trade-off.

Comparison with different PEFT methods. Table **??** provides a comprehensive comparison between our method and representative parameter-efficient fine-tuning (PEFT) baselines in terms of segmentation accuracy (mIoU), training efficiency, and resource consumption. Although LoRA achieves competitive training speed (33 s/epoch), it suffers from performance degradation across all label budgets, underperforming the full fine-tuning baseline by 1.34, 1.35, and 1.53 mIoU points on the 1/16, 1/8, and 1/4 splits, respectively—suggesting that simple low-rank adaptation is insufficient to capture the complex multi-scale structures inherent in remote sensing imagery. Similarly, SSF and BitFit further reduce trainable parameters but incur larger accuracy drops, indicating that overly aggressive parameter reduction compromises representational capacity. Conv_LoRA struggles to improve accuracy and remains relatively slow (64 s/epoch), demonstrating that naively incorporating convolutional inductive biases without careful architectural design yields limited benefits. HydraLoRA offers a competitive speed–accuracy trade-off among prior PEFT methods (38 s/epoch) but incurs the largest memory footprint (24,527 MB), making it impractical for resource-constrained scenarios. In contrast, our method achieves a favorable trade-off across multiple dimensions: it delivers competitive accuracy (76.97, 77.64, and 77.82 mIoU) while requiring only 10.38M trainable parameters—a substantial reduction from the 102.4M parameters in full fine-tuning. Moreover, SemiFT attains a competitive training speed (32 s/epoch) and utilizes the lowest peak memory (8,427 MB) among all methods, demonstrating its practical efficiency. These results validate that our approach effectively balances accuracy and efficiency, enabling robust performance under limited supervision without imposing prohibitive computational costs.

Method	Memory	Train speed	Param	1/16	1/8	1/4
Baseline	11710 MB	47s/epoch	102.4M	78.29	78.25	78.68
LoRA	9895 MB	33s/epoch	10.67M	76.95	76.90	77.15
SSF	8089 MB	31s/epoch	9.68M	75.07	75.10	75.19
bitfit	7768 MB	31s/epoch	9.75M	74.75	75.39	75.89
AdaptFormer	8099 MB	31s/epoch	10.23M	75.27	75.91	76.20
FacTT_TT	9433 MB	45s/epoch	10.38M	76.65	77.27	77.08
FacTT_TK	9413 MB	35s/epoch	10.63M	76.85	76.64	77.48
Conv_LoRA	17263 MB	64s/epoch	16.44M	74.34	74.85	74.16
HydraLoRA	24527 MB	38s/epoch	12.66M	77.26	76.96	77.35
Ours	8427 MB	32s/epoch	10.38M	76.97	77.64	77.82

Table 4: Comparison with parameter-efficient fine-tuning (PEFT) baselines on the Vaihingen dataset under different labeled-data splits.Memory denotes peak GPU memory consumption.

Different Scale and Angle Factors. Table **??** systematically investigates the effect of the rotation range θ and the scale set s in RGCR under the 1/4 split. Using no augmentation ($\theta = 0$, $s = 1$) yields a baseline performance of 78.25 mIoU. Introducing a small set of discrete rotations provides marginal gains (78.46 mIoU, +0.21), whereas sampling rotations from the full range ($0 \sim 360$) substantially improves performance to 78.77 mIoU (+0.52), indicating that strong rotational diversity better regularizes the model and enhances invariance to orientation variations. Using a coarse discrete grid (step=30) performs slightly worse (78.67 mIoU, -0.10 compared to continuous sampling), suggesting that denser rotation sampling

θ **(rotation)**	s **(scale)**	**1/8**
0	$\{1\}$	78.25
$\{-15, -10, -5, 0, 5, 10, 15\}$	$\{1\}$	78.46
$0 \sim 360$	$\{1\}$	78.77
$0 \sim 360$ (step=30)	$\{1\}$	78.67
$0 \sim 360$	$\{1.0, 1.25, 1.5, 2.0\}$	78.78
$0 \sim 360$	$\{0.5, 0.75, 1.0\}$	79.16
$0 \sim 360$	$0.5 \sim 2.0$	79.25
$0 \sim 360$	$\{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$	**79.29**

Table 5: Hyper-parameter study of the proposed RGCR module on the Vaihingen dataset under the 1/8 labeled-data split.

is beneficial for capturing fine-grained rotational patterns. We further study the interaction between rotation and scale augmentation. When combining full-range rotations with only up-scaling factors ($s = 1.0, 1.25, 1.5, 2.0$), performance reaches 78.78 mIoU, providing only marginal improvement over rotation alone. In contrast, incorporating down-scaling ($s = 0.5, 0.75, 1.0$) significantly improves results to 79.16 mIoU (+0.39), implying that excluding down-scaling may bias the model toward enlarged views and weaken robustness to scale variations. The best performance is achieved when using full-range rotations together with a comprehensive multi-scale set, reaching 79.29 mIoU with $s = 0.5, 0.75, 1.0, 1.25, 1.5, 2.0$ (and 79.25 mIoU with the continuous range $0.5 \sim 2.0$). Overall, these results demonstrate that RGCR benefits from rich rotation augmentation and balanced multi-scale sampling, with particular emphasis on including smaller scales to improve robustness to viewpoint and scale variations inherent in remote sensing imagery.

5 Conclusion

In this work, we explore semi-supervised fine-tuning, aiming to adapt VFMs with scarce pixel-level labels and minimal trainable parameters. This calls for improved multi-scale context modeling under a strict parameter budget, together with stable exploitation of unlabeled imagery without full-model updates. To this end, we propose SemiFT, a semi-supervised fine-tuning framework that decouples adaptation into (i) a Scale-Aware Mixture-of-Experts (SA-MoE) for scale-specific specialization and dynamic collaboration, and (ii) rotation-based geometric-invariant consistency regularization (RGCR) to inject geometric priors and enhance orientation robustness. Experiments under scarce annotations verify that SemiFT improves accuracy with far fewer trainable parameters, offering an excellent accuracy–efficiency balance and stronger adaptation. Future work will extend this paradigm to broader remote-sensing tasks and explore more efficient expert routing and stronger unlabeled supervision.